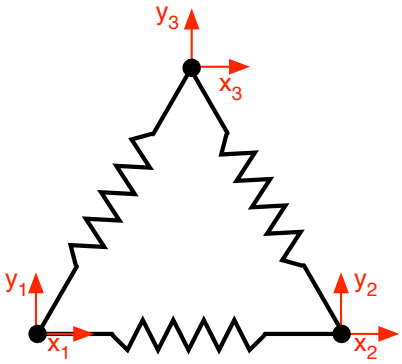
Vibrations of Ozone

Ozone (O_3) is a tri-atomic molecule with the shape of an equilateral triangle. Out of lack of inspiration, we will use six cartesian degrees of freedom to describe the dynamics, even through there clearly are translational and rotational symmetries in the problem that should inspire us to reduce the problem to three generalized coordinates. As we will see, we'll end up with zero frequency modes.



Normal mode analysis of vibrations of Ozone

The potential energy is only non-zero when springs are stretched or compressed; that is, when the masses move with the projection of their displacements *along* a spring. In terms of the coordinates

$$\mathbf{q} = (x_1, y_1, x_2, y_2, x_3, y_3),$$

the potential energy is:

$$V = \tfrac{1}{2}K\left\{\left[\tfrac{1}{2}(x_2 - x_3) + \tfrac{\sqrt{3}}{2}(y_3 - y_2)\right]^2 + \left[\tfrac{1}{2}(x_3 - x_1) + \tfrac{\sqrt{3}}{2}(y_3 - y_1)\right]^2 + (x_2 - x_1)^2\right\}$$

and the *matrix* V_{ij} is

$$V_{14} = \frac{\partial^2 V}{\partial x_1 \partial y_2} \quad \text{etc.}$$

$$\mathbf{V} = K\begin{pmatrix} 5/4 & \sqrt{3}/4 & -1 & 0 & -1/4 & -\sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 & 0 & 0 & -\sqrt{3}/4 & -3/4 \\ -1 & 0 & 5/4 & -\sqrt{3}/4 & -1/4 & \sqrt{3}/4 \\ 0 & 0 & -\sqrt{3}/4 & 3/4 & \sqrt{3}/4 & -3/4 \\ -1/4 & -\sqrt{3}/4 & -1/4 & \sqrt{3}/4 & 1/2 & 0 \\ -\sqrt{3}/4 & -3/4 & \sqrt{3}/4 & -3/4 & 0 & 3/2 \end{pmatrix}$$

$$\mathbf{T} = m\mathbf{I}$$

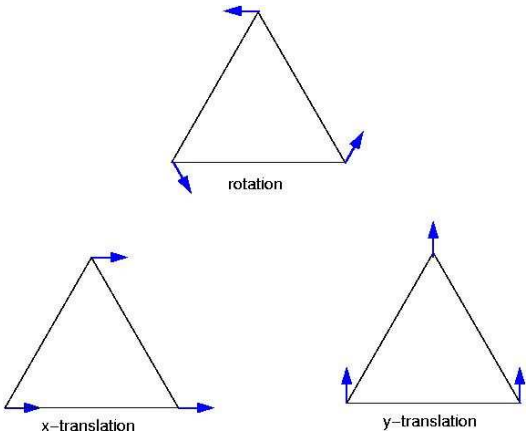
The kinetic energy matrix is $T = mI$, where I is the 6×6 unit matrix in the $\mathbf{q}$ basis. Linear algebra with Eq. (6) gives the normal mode eigenvalues ω_α^2 and eigenvectors:

$$\text{Eigenvalues}[\mathbf{V}/m]: \quad \left\{0, 0, 0, \frac{3k}{2m}, \frac{3k}{2m}, \frac{3k}{m}\right\}$$

The general motion is a linear superposition of the 6 normal eigenmodes, with coefficients that are the *normal mode coordinates*.

Zero frequency modes

How do we identify the eigenvectors associated with the three degenerate zero frequency modes? We can invoke symmetries to convince ourselves that they correspond to two translational and one rotational degrees of freedom (in the 2-D $x-y$ plane). These have to be mutually orthogonal eigenvectors (convince yourself that they are!):



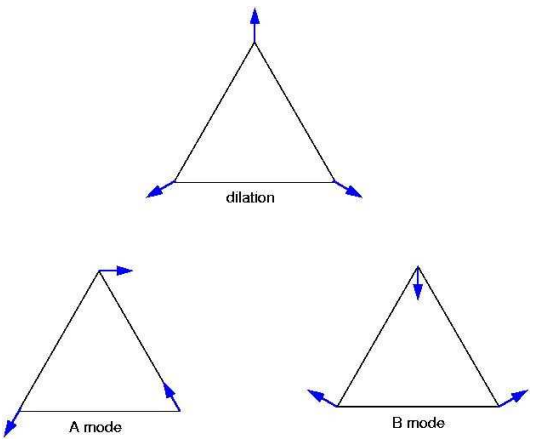
Vibrational modes

The three other modes are vibrational. Standard but tedious linear algebra (use MATLAB or Mathematica) gives us the orthonormal eigenvectors (two of which are degenerate):

$$\text{Eigenvectors[V][[6]]:} \quad \left\{-\frac{\sqrt{3}}{2}, -\frac{1}{2}, \frac{\sqrt{3}}{2}, -\frac{1}{2}, 0, 1\right\}$$

$$\text{Eigenvectors[V][[5]]:} \quad \left\{-\frac{1}{2}, -\frac{\sqrt{3}}{2}, -\frac{1}{2}, \frac{\sqrt{3}}{2}, 1, 0\right\}$$

$$\text{Eigenvectors[V][[4]]:} \quad \left\{\frac{-\sqrt{3}}{2}, \frac{1}{2}, \frac{\sqrt{3}}{2}, \frac{1}{2}, 0, -1\right\}$$



A and B modes are *degenerate*. These solutions are the generators of symmetry transformations (rotations about 120°), as described by group theory for $\bar{6}m2$ point group.